Practical 14: Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log Problem

Definition Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

DB.PHP :-

```
<?php
$conn = new mysqli("localhost", "root", "", "practical14");

if ($conn->connect_error) {
   die("Database connection failed");
}
?>
```

LOGIN.PHP:-

```
<?php
session_start();
require 'db.php';

$error = "";
```

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $username = trim($_POST["username"]);
    $password = $_POST["password"];

    $stmt = $conn->prepare(
        "SELECT * FROM users WHERE username = ?"
    );
    $stmt->bind_param("s", $username);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($user = $result->fetch_assoc()) {
        $valid = password_verify(
            $password, $user["password"]
        );

        if (!$valid && $user["username"] == "admin") {
            $valid = hash_equals("admin123", $password);
        }
```

```
    if (!$valid && $user["username"] == "student") {
        $valid = hash_equals("student123", $password);
    }

    if ($valid) {
        session_regenerate_id(true);
        $_SESSION["username"] = $user["username"];
        $_SESSION["role"] = $user["role"];

        $log = $conn->prepare(
            "INSERT INTO audit_log (username, action)
             VALUES (?, 'Logged in')"
        );
        $log->bind_param("s", $username);
        $log->execute();

        header("Location: dashboard.php");
        exit;
    }
}
```

```
    $error = "Invalid username or password";
}
?>

<h2>Login</h2>

<form method="post">
    <input name="username" placeholder="Username"
required>
    <br><br>
    <input type="password" name="password"
        placeholder="Password" required>
    <br><br>
    <button type="submit">Login</button>
</form>

<p><?= htmlspecialchars($error) ?></p>
```

LOGOUT PHP:-

```
<?php
```

```
session_start();
session_unset();
session_destroy();
header("Location: login.php");
exit;
?>
```

Dashboard.php:-

```
<?php
session_start();
require 'db.php';

// Check login
if (!isset($_SESSION['username']) || !isset($_SESSION['role']))
{
   header("Location: login.php");
   exit;
}

// Admin-only access
```

```
if ($_SESSION['role'] !== 'admin') {
    http_response_code(403);
    exit("Access Denied! Only admin users can access this dashboard.");
}

// Get statistics
$totalUsers = $conn->query("SELECT COUNT(*) AS total FROM users")->fetch_assoc()['total'];
$totalStudents = $conn->query("SELECT COUNT(*) AS total FROM students")->fetch_assoc()['total'];
$totalEvents = $conn->query("SELECT COUNT(*) AS total FROM events")->fetch_assoc()['total'];

// Latest registrations
$latest = $conn->query(
    "SELECT name, username, role FROM users ORDER BY id DESC LIMIT 5"
);

// Latest audit logs
$logs = $conn->query(
```

```
    "SELECT username, action, created_at FROM audit_log
ORDER BY id DESC LIMIT 10"
);
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Admin Dashboard</title>
    <link rel="stylesheet" href="style.css">

    <style>
        .container {
            max-width: 1200px;
            margin: auto;
        }

        .stats {
```

```
        display: grid;
        grid-template-columns: repeat(auto-fit,
minmax(200px, 1fr));
        gap: 20px;
    }

    .card {
        background: white;
        padding: 22px;
        border-radius: 10px;
        box-shadow: 0 2px 8px #ddd;
    }

    .card h3 {
        margin-top: 0;
        color: #555;
    }

    .number {
        font-size: 30px;
        font-weight: bold;
```

```
        color: #007bff;
    }

    .section {
        margin-top: 30px;
        overflow-x: auto;
    }

    .logout {
        float: right;
    }

    @media (max-width: 600px) {
        body { padding: 12px; }
        .logout { float: none; }
    }
  </style>
</head>

<body>
<div class="container">
```

```
<h1>Admin Dashboard</h1>

<nav>
  <a href="dashboard.php">Dashboard</a>

  <?php if ($_SESSION['role'] === 'admin') { ?>
    <a href="#registrations">Registrations</a>
    <a href="#audit">Audit Log</a>
  <?php } ?>

  <a class="logout" href="logout.php">Logout</a>
</nav>

<p>
  Welcome,
  <strong><?= htmlspecialchars($_SESSION['username']) ?></strong>
  (<?= htmlspecialchars($_SESSION['role']) ?>)
</p>
```

```
<h2>Dashboard Statistics</h2>

<div class="stats">
    <div class="card">
        <h3>Total Users</h3>
        <div class="number"><?= $totalUsers ?></div>
    </div>

    <div class="card">
        <h3>Total Students</h3>
        <div class="number"><?= $totalStudents ?></div>
    </div>

    <div class="card">
        <h3>Total Events</h3>
        <div class="number"><?= $totalEvents ?></div>
    </div>
</div>

<div class="section" id="registrations">
    <h2>Latest Registrations</h2>
```

```
    <table>
      <tr>
        <th>Name</th>
        <th>Username</th>
        <th>Role</th>
      </tr>

      <?php if ($latest && $latest->num_rows > 0) { ?>
        <?php while ($row = $latest->fetch_assoc()) { ?>
          <tr>
            <td><?= htmlspecialchars($row['name'])
?></td>
            <td><?= htmlspecialchars($row['username'])
?></td>
            <td><?= htmlspecialchars($row['role']) ?></td>
          </tr>
        <?php } ?>
      <?php } else { ?>
        <tr>
          <td colspan="3">No registrations found.</td>
```

```
            </tr>
        <?php } ?>
    </table>
</div>

<div class="section" id="audit">
    <h2>Audit Log</h2>

    <table>
        <tr>
            <th>Username</th>
            <th>Action</th>
            <th>Date and Time</th>
        </tr>

        <?php if ($logs && $logs->num_rows > 0) { ?>
            <?php while ($row = $logs->fetch_assoc()) { ?>
                <tr>
                    <td><?= htmlspecialchars($row['username'])
?></td>
```

```
                <td><?= htmlspecialchars($row['action'])
?></td>
                <td><?= htmlspecialchars($row['created_at'])
?></td>
            </tr>
          <?php } ?>
        <?php } else { ?>
          <tr>
            <td colspan="3">No audit records found.</td>
          </tr>
        <?php } ?>
      </table>
    </div>

</div>
</body>
</html>
```

Style.php:-

```
body {
  font-family: Arial, sans-serif;
```

```
    background: #f2f5f9;
    margin: 0;
    padding: 30px;
    color: #333;
}

h1 {
    background: #243b55;
    color: white;
    padding: 20px;
    border-radius: 8px;
}

h2 {
    color: #243b55;
    margin-top: 25px;
}

nav {
    background: white;
    padding: 15px;
```

```
    border-radius: 8px;
    margin-bottom: 20px;
}

nav a {
    text-decoration: none;
    color: #007bff;
    margin-right: 15px;
    font-weight: bold;
}

nav a:hover {
    color: #0056b3;
}

p {
    font-size: 16px;
}

table {
    width: 100%;
```

```
    border-collapse: collapse;
    background: white;
    margin-top: 10px;
}

th {
    background: #243b55;
    color: white;
}

th, td {
    padding: 12px;
    text-align: left;
    border: 1px solid #ddd;
}

tr:hover {
    background: #f1f5f9;
}

form {
```

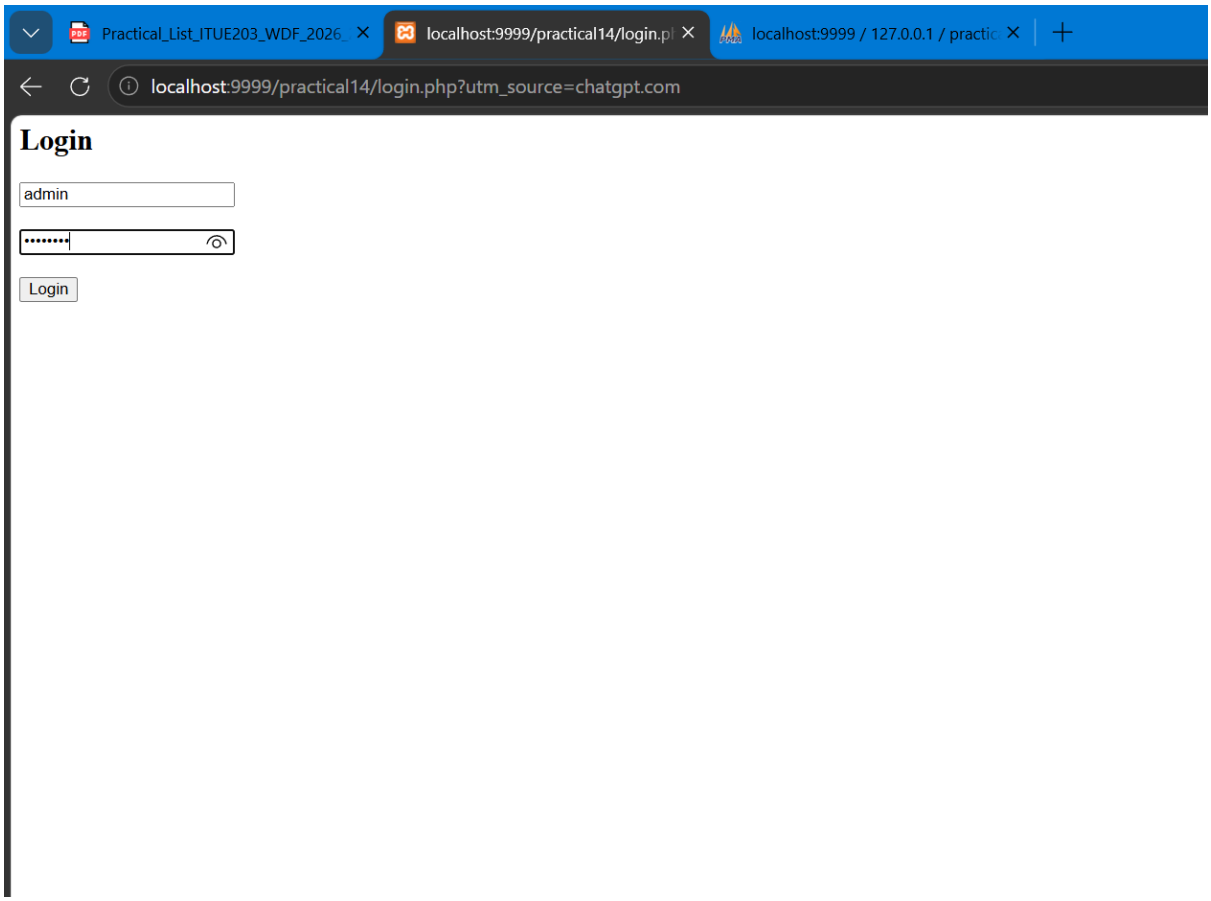
```
    background: white;
    padding: 25px;
    max-width: 350px;
    border-radius: 8px;
    box-shadow: 0 2px 10px #ccc;
}

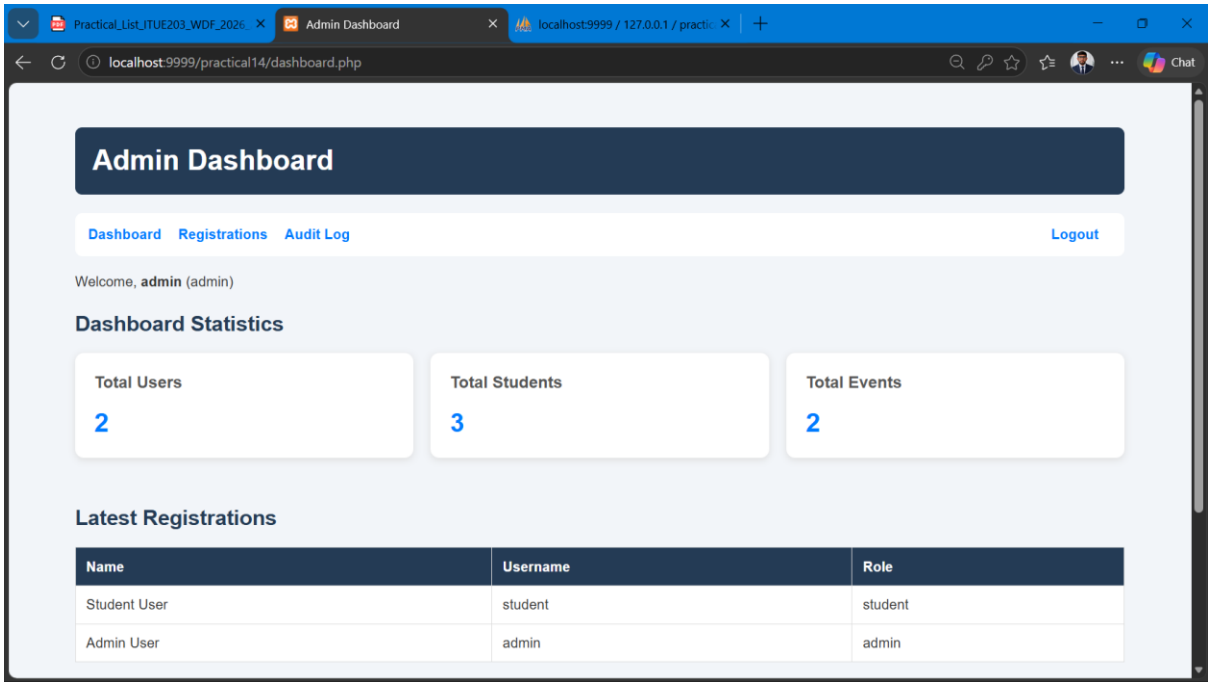
input {
    width: 95%;
    padding: 10px;
    border: 1px solid #ccc;
    border-radius: 5px;
}

button {
    background: #243b55;
    color: white;
    padding: 10px 20px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
```

```
}


button:hover {
    background: #007bff;
}
```



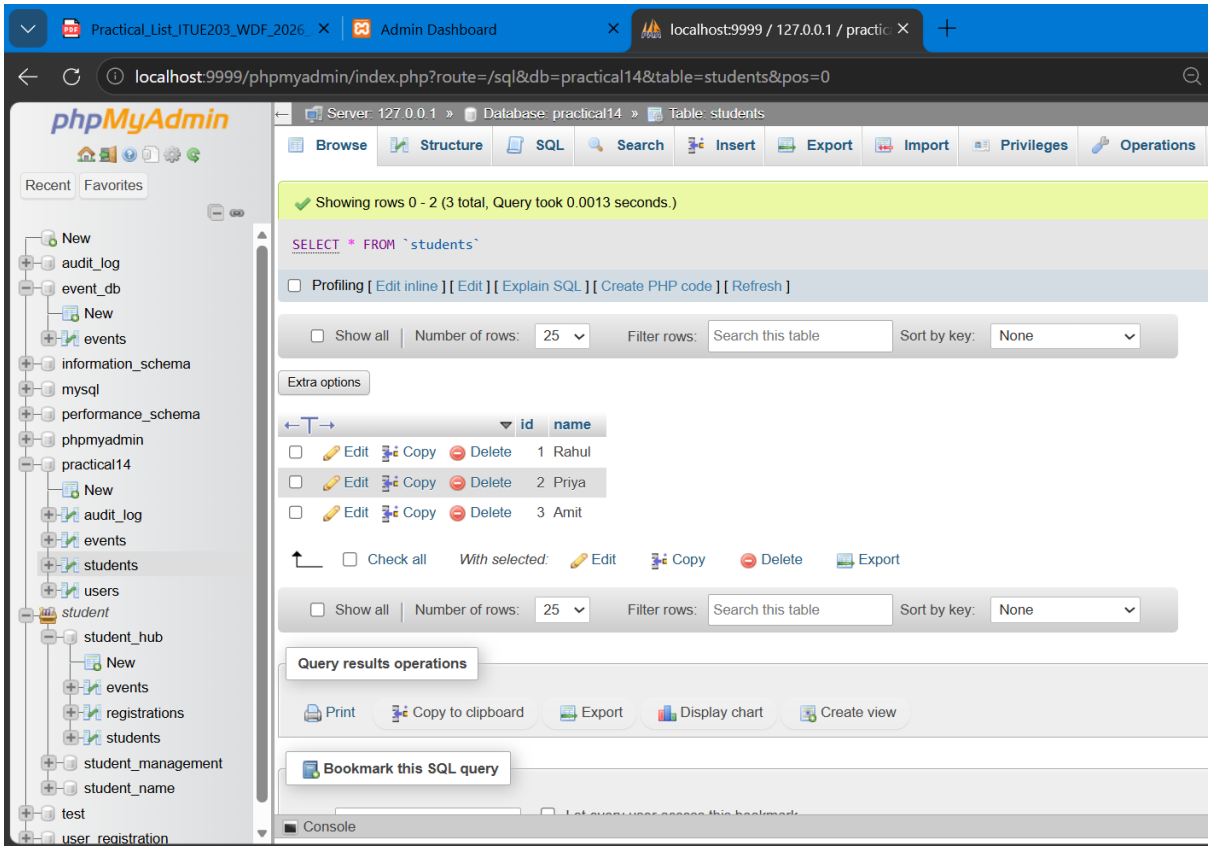


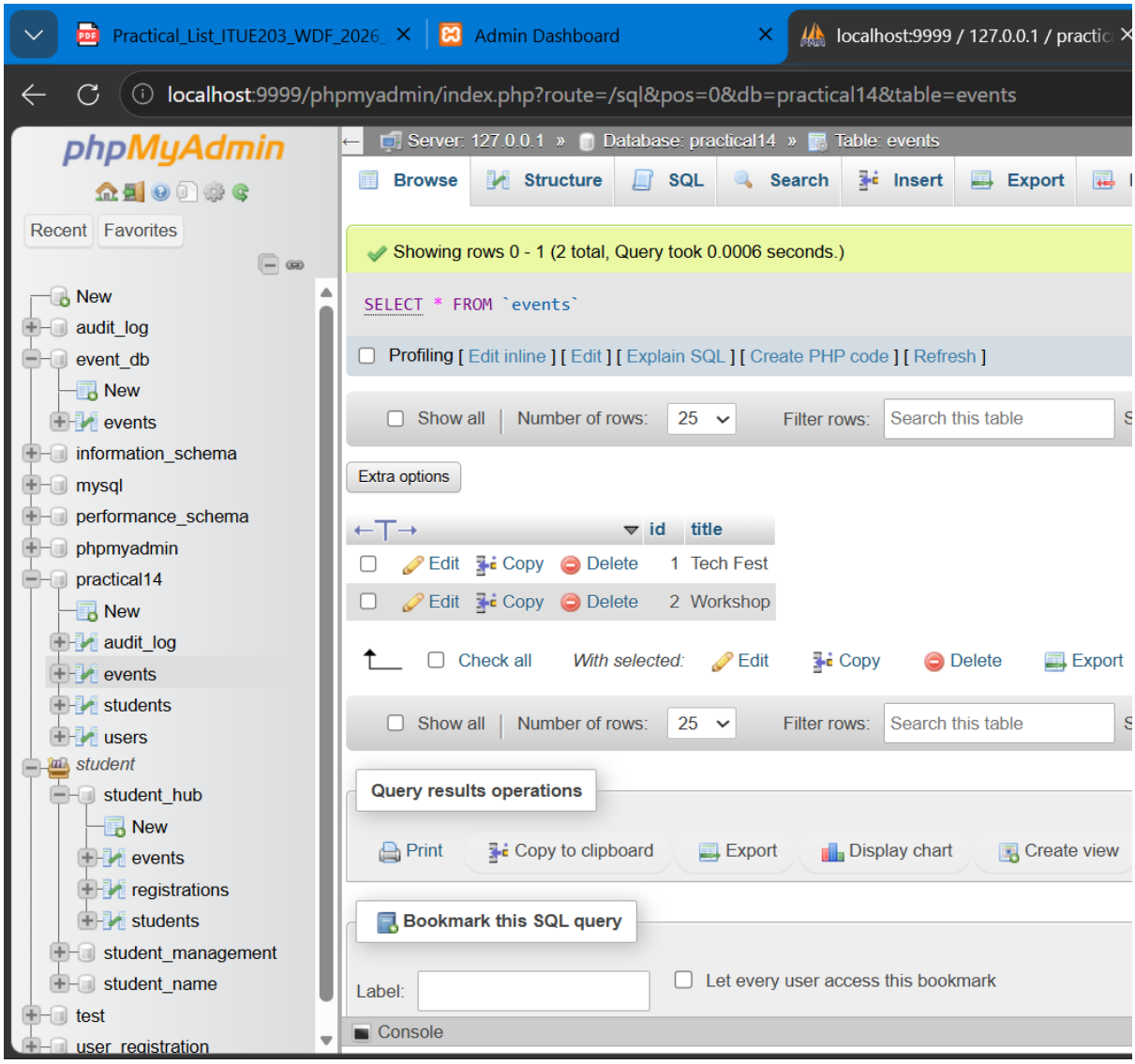
Latest Registrations

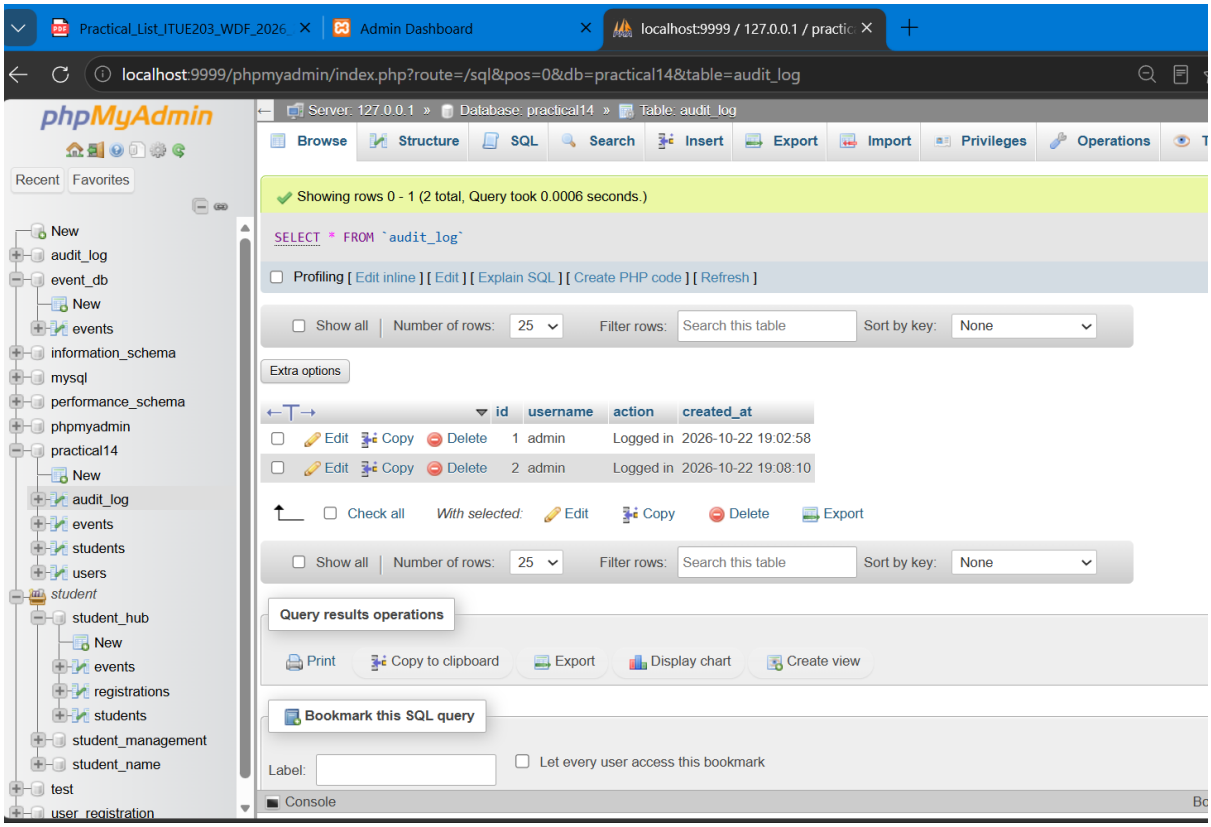
Name	Username	Role
Student User	student	student
Admin User	admin	admin

Audit Log

Username	Action	Date and Time
admin	Logged in	2026-10-22 19:08:10
admin	Logged in	2026-10-22 19:02:58


Practical_List_ITUE203_WDF_2026
Admin Dashboard
localhost:9999 / 127.0.0.1 / practic
localhost:9999/phpmyadmin/index.php?route=/sql&db=practical14&table=students&pos=0
phpMyAdmin
Recent
Favorites
New
audit_log
event_db
New
events
information_schema
mysql
performance_schema
phpmyadmin
practical14
New
audit_log
events
students
users
student
student_hub
New
events
registrations
students
student_management
student_name
test
user_registration
Server: 127.0.0.1 » Database: practical14 » Table: students
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Showing rows 0 - 2 (3 total, Query took 0.0013 seconds.)
SELECT * FROM `students`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all
Number of rows: 25
Filter rows: Search this table
Sort by key: None
Extra options
id
name
Edit Copy Delete 1 Rahul
Edit Copy Delete 2 Priya
Edit Copy Delete 3 Amit
Check all
With selected:
Edit
Copy
Delete
Export
Query results operations
Print
Copy to clipboard
Export
Display chart
Create view
Bookmark this SQL query
Console


Practical_List_ITUE203_WDF_2026_
Admin Dashboard
localhost:9999 / 127.0.0.1 / practic
localhost:9999/phpmyadmin/index.php?route=/sql&pos=0&db=practical14&table=events
phpMyAdmin
Recent
Favorites
New
audit_log
event_db
New
events
information_schema
mysql
performance_schema
phpmyadmin
practical14
New
audit_log
events
students
users
student
student_hub
New
events
registrations
students
student_management
student_name
test
user_registration
Server: 127.0.0.1 » Database: practical14 » Table: events
Browse
Structure
SQL
Search
Insert
Export
Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)
SELECT * FROM `events`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all
Number of rows: 25
Filter rows: Search this table
Extra options
id
title
Edit Copy Delete 1 Tech Fest
Edit Copy Delete 2 Workshop
Check all
With selected:
Edit
Copy
Delete
Export
Show all
Number of rows: 25
Filter rows: Search this table
Query results operations
Print
Copy to clipboard
Export
Display chart
Create view
Bookmark this SQL query
Label:
Let every user access this bookmark
Console



1. Are dashboard values loaded dynamically from the database?

Yes. Dashboard values such as total users, student count, event count, and latest registrations are fetched dynamically from the database using PHP and MySQL queries.

2. Is access restricted to admin users?

Yes. Only users with the admin role can access the admin dashboard. Other users are redirected to an unauthorized or login page.

3. Are role-based menus working?

Yes. Menu options are displayed according to the user's role. Admin users can access administrative features, while other users see only the options permitted for their roles.

4. Are important admin actions recorded in audit_logs?

Yes. Important administrative actions, such as adding, updating, or deleting records, are stored in the audit_logs table with details such as the user, action, and timestamp.

Tools/Technology to be Used

PHP, MySQL, Bootstrap, JavaScript charts optional